



SUSTAINABILITY & CIRCULAR METALLURGY

Industrial Circularity: Closed-Loop Scrap Recycling & Decarbonization

Integrating electric induction furnace remelting, secondary alloy recovery, and Scope 1/2 emissions reduction across heavy manufacturing operations.

Dawit Tadesse

Chief Operating Officer // Asterra Manufacturing Group S.C.

EXECUTIVE SUMMARY

Heavy industry and metals fabrication are historically carbon-intensive sectors. As Asterra expanded to 120,000 MT annual processing capacity across four plant complexes, corporate leadership mandated a closed-loop scrap recycling model. By capturing internal plate off-cuts, re-melting in computerized electric induction furnaces powered by Ethiopia's renewable hydroelectric grid, and recovering high-value alloying elements, Asterra has decoupled production growth from greenhouse gas emissions.

Key Performance Findings & Metrics

- **94.2% Internal Scrap Capture:** High-efficiency automated collection systems throughout CNC plasma profiling, laser cutting, and beam profiling facilities routed 22,400 metric tons of clean scrap directly back to secondary remelting.
- **62% Embodied Carbon Reduction:** Utilizing secondary recycled scrap in electric induction furnaces achieves an audited 62% reduction in Scope 1 & 2 carbon emissions per ton of finished structural component compared to primary blast furnace ironmaking.
- **98.5% Alloying Element Recovery:** Precision slag treatment and chemical optical spectrometry enable near-complete reclamation of manganese, silicon, and vanadium additions.

1. Industrial Responsibility at Scale

True industrial authority demands rigorous environmental stewardship. Rather than exporting scrap metal as low-value raw waste, Asterra has established an integrated secondary metallurgy cycle in Industrial Zone 4. Scrap generated during heavy beam fabrication and UHPC mold assembly is categorized by metallurgical grade and directly reused in secondary structural cast components.

2. Clean Hydroelectric Grid Integration

Operating in Ethiopia provides Asterra with a strategic decarbonization asset: over 95% of grid electricity is derived from clean hydroelectric and renewable energy generation. By shifting from fossil-fuel-fired reheating to high-efficiency electric induction coils, Asterra maintains one of the lowest carbon footprints per ton of fabricated structural steel in East Africa.

ENVIRONMENTAL METRIC	BASELINE (2020)	2024 AUDITED PERFORMANCE	ISO 14001:2015 TARGET
Internal Scrap Recovery Rate	76.4%	94.2%	≥ 90.0%
Scope 1 & 2 Emissions per MT Steel	1.82 tCO ₂ e	0.69 tCO ₂ e (-62%)	≤ 0.75 tCO ₂ e
Renewable Energy Share in Production	88.0%	98.5%	100.0% by 2028
Industrial Wastewater Recirculation	64.0%	91.2%	≥ 90.0%
Hazardous Waste Diversion Rate	92.5%	99.4%	100.0%

3. Circularity Beyond Steel: Polymer & Aggregate Recycling

In Division 03 (Polymers) and Division 02 (Precast), identical circular principles govern operations. Polymer extrusions that fail dimensional laser telemetry are granulated and re-compounded into non-pressure agricultural conduits. Concrete wash water and aggregate fines are filtered and recycled back into commercial mortar batching.